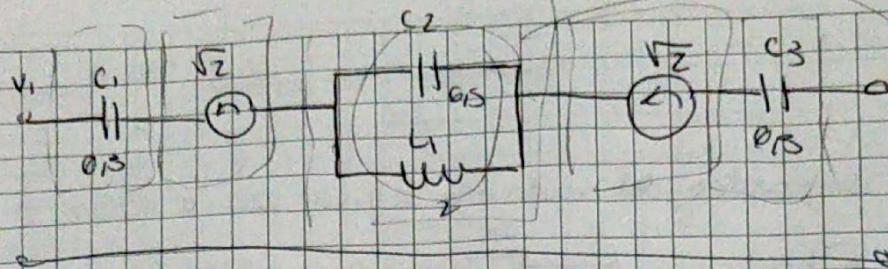


6



$$V_s = V_{C1} + V_{L1} = \frac{1}{s} C_1 + \frac{1}{s L_1} = \frac{s^2 C_1 L_1 + 1}{s L_1} = \frac{s^2 + 1}{s L_1}$$

Circuito lattice equivalente

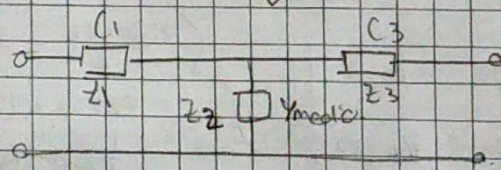
$$T_{final} = T_{C1} \cdot T_{L1} \cdot T_{rs} \cdot T_{L2} \cdot T_{C3}$$

$$T_{final} = \begin{bmatrix} 1 & \frac{s}{2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & \sqrt{2} \\ \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} 1 & \frac{2s}{s^2+1} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & \sqrt{2} \\ \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} 1 & \frac{s}{2} \\ 0 & 1 \end{bmatrix}$$

pero por la par de un T a un lattice es fácil
que sea una Y.

$$T_{medoy} = \begin{bmatrix} 0 & \sqrt{2} \\ \frac{1}{\sqrt{2}} & \frac{s^2+1}{s} \end{bmatrix} \begin{bmatrix} 0 & \sqrt{2} \\ \frac{1}{\sqrt{2}} & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{2s}{s^2+1} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{s}{s^2+1} & 1 \end{bmatrix}$$

en el girador pasamos de una Z a una Y



$$Z_{final} = \begin{bmatrix} Z_1 + Z_2 & Z_2 \\ Z_2 & Z_2 + Z_3 \end{bmatrix}$$

$$\rightarrow Z_1 + Z_2 = \frac{2}{s} + \frac{s^2+1}{s} \rightarrow Z_2 = \frac{s^2+1}{s}$$

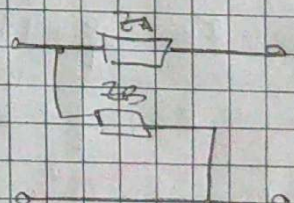
$$\rightarrow Z_2 + Z_3 = \frac{s^2+1}{s} + \frac{2}{s}$$

$Z_1 = Z_3$

$$Z_{final} = \begin{bmatrix} \frac{Z_1+Z_2}{2} & \frac{Z_2-Z_1}{2} \\ \frac{Z_2-Z_1}{2} & \frac{Z_2+Z_3}{2} \end{bmatrix}$$

$$\rightarrow \frac{2}{s} + \frac{s^2+1}{s} = \frac{Z_A + Z_B}{2} \rightarrow \frac{s^2+3}{s} = \frac{Z_A + Z_B}{2}$$

$$\rightarrow \frac{s^2+1}{s} = \frac{Z_B - Z_A}{2} \rightarrow \frac{(s^2+1)2 + Z_A - Z_B}{s} = 2Z_B$$



$$\frac{(s^2+3)2}{s} = Z_A + \frac{(s^2+1)2 + Z_A}{s}$$

$$\frac{2s^2+6}{s} - \frac{(s^2+1)2}{s} = Z_A$$

$$\frac{4}{s} = 2Z_A$$

$$\frac{2}{s} = Z_A$$

$$\frac{(s^2+1)2 + 2}{s} = Z_B$$

$$\frac{2s^2+2+2}{s} = Z_B$$

$$\frac{2(s^2+2)}{s} = Z_B$$

$$2s + \frac{4}{s} = Z_B$$

